CHAPTER 1

Numerical sequences and discrete models

Chapter overview

Objectives. Move from narrative to a sequence; distinguish increment and rate; calculate terms and sums; justify a variation; experiment a threshold.
Prerequisites. Literal calculation, integer powers, percentages, affine function, Python loop.
Annual benchmark. 8·17 September: 8 hours, then review

Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. Calculate $3 + 2 \times 4$.

2. Calculate 2^0, 2^3 and 2^{-2}.

3. A staff strength of 80 is increasing by 5 per cent.

4. For $f(x) = 3x - 2$, calculate $f(0)$ and $f(4)$.

5. How many integers are there from 3 to 10, including limits?

6. A loop `for k in range(4)` takes what values of k ?

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. In 90 seconds, explain the difference between rank, term and number of terms. Give a counter-example to a guess rule on three values.

Search before formalizing

In pairs, compare two collection offers: 20 kg the first week then 5 kg each
20 kg and then a 20% increase each week. Conjecture which collects the most at
All in six weeks, then justify. Allow 10 min of research and 5 min of confrontation.

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min.
Keep a trace of a test, even if not completed. A conjecture must then be justified.

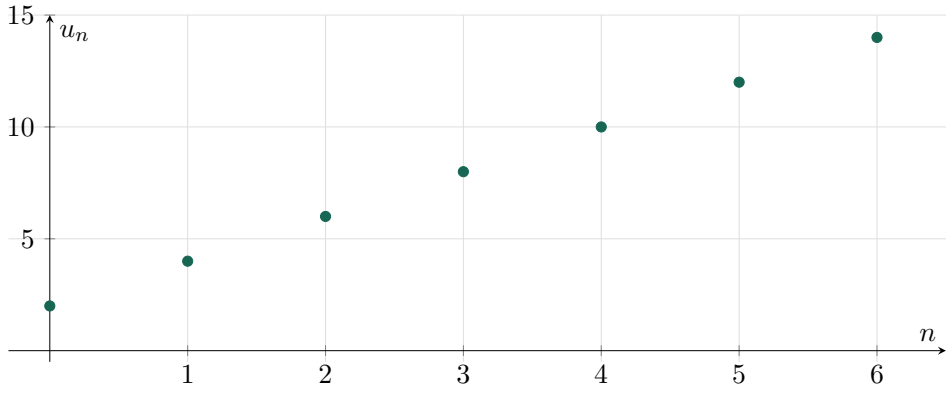
1.1 Defining and Representing a Succession

Definition and results

A real sequence is a function whose variable is an integer: at each row n authorized corresponds to a real u_n. Rating (u_n) means the continuation, u_n a term. An explicit formula gives directly u_n according to the n. A definition by induction provides a first term and a rule of passage. Representation is a set of points ($n; u_n$) : one does not connect necessarily these points.

Status and justification. Definition. The uniqueness of terms obtained from a first term and a deterministic rule is accepted; no formal induction reasoning is required.

Example solved. For $u_n = n^2 + 1$, $u_0 = 1$, $u_3 = 10$. For $v_0 = 1$ and $v_{n+1} = 2v_n + 1$, $v_1 = 3$, $v_2 = 7$, $v_3 = 15$. We can't replace v_3 by 2 $\times 3 + 1$: the rule relates to the previous term.



Terms of $u_n = 2n + 2$: only the integer rows are represented.

Point of vigilance

The row is not the value. A finished list of terms does not determine a single rule.

Immediate application. Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; propose an explicit formula.

1.2 Arithmetic Sequences

Definition and results

A sequence is arithmetical of common difference r if $u_{n+1} = u_n + r$ for any authorized rank. $u_n = u_p + (n-p)r$ for $n \geq p$. If $r > 0$, it is strictly increasing; if $r < 0$, strictly Decreasing; if $r = 0$, constant. Points are aligned on a right.

Status and justification. For p fixed, the continuation $v_n = u_p + (n-p)r$ checks $v_p = u_p$ and $v_{n+1} = v_n + r$. By the uniqueness of successive terms, $u_n = v_n$. Variation is the result of the $u_{n+1} - u_n = r$.

Example solved. A reserve contains 140 L and receives 12 L per minute. n minutes, $u_n = 140+12n$. At 5 min: 200 L. If we start indexing at 1 with 140 L, we write $u_n = 140+12(n-1)$.

Point of vigilance

Adding a constant percentage does not usually give an arithmetic sequence.

Immediate application. We know. $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. r and u_0.

1.3 Geometric sequences

Definition and results

A sequence is geometrical of common ratio q if $u_{n+1} = qu_n$. For $n \geq p$, $u_n = u_p q^{n-p}$ with the agreement that an empty product is worth 1. A constant rate t gives $q = 1 + t$. Variation decides with $u_{n+1} - u_n = (q-1)u_n$. For $q < 0$, the signs may alternate.

Status and justification. The formula gives the first term and its multiplication by q gives the next term. The uniqueness of the generation justifies the formula. $q = 0$, all terms after first are null; deal with this case directly.

Example solved. A quantity of 160 mg loses 25% at each step: $u_n = 160(\frac{3}{4})^n$. Terms are positive and $u_{n+1} - u_n = -\frac{u_n}{4} < 0$. We get $u_2 = 90$ mg, not to be confused 25% decrease and 25 mg withdrawal.

Point of vigilance

$q > 1$ is not enough to conclude that any geometric sequence is increasing: the sign of the first term counts.

Immediate application. Studying the variation in $u_n = -2 \times 3^n$ and then $v_n = 2(-1)^n$.

1.4 Amounts, indices and demonstrations

Definition and results

For $n \geq 1$, $1+2+\cdots+n = \dfrac{n(n+1)}{2}$. For $q \neq 1$ and $n \geq 0$, $1+q+\cdots+q^n = \dfrac{1-q^{n+1}}{1-q}$. For $q = 1$, the sum is $n+1$. Thus, u_p to u_n, an arithmetic sum is $\dfrac{(n-p+1)(u_p+u_n)}{2}$; a geometric sum is $\dfrac{u_p(1-q^{n-p+1})}{1-q}$ if $q \neq 1$.

Status and justification. Writing $S = 1 + \cdots + n$ in both directions, each pair of the sum 2 S value $n+1$, with n pairs. $T = 1 + q + \cdots + q^n$, subtract $qT = q + \cdots + q^{n+1}$: $(1-q)T = 1 - q^{n+1}$. The case $q = 1$ must precede division.

Example solved. For $u_n = 3 + 2n$, $u_2 + \cdots + u_6 = \dfrac{5(7+15)}{2}$ = 55. For $v_n = 2 \times 3^n$, $v_0 + \cdots + v_3 = \dfrac{2(1-3^4)}{1-3} = 80$.

Point of vigilance

Count the terms before using the formula: from 0 to n, there are n+1.

Immediate application. Calculate $\sum_{k=2}^{5} 2^k$ without summation symbol in the answer.

1.5 Conjecture a limit and schedule a threshold

Definition and results

A limit is discussed here intuitively. $\dfrac{1}{n+1}$ close to 0; n^2 Grows without limit; $(-1)^n$ a digital experience suggests a conclusion but does not prove behavior for all ranks. A threshold algorithm must specify the a sense of inequality and provision for a stop.

Status and justification. Illustrative findings, without formal theory of limits. integer threshold is verified by checking the row found and the previous row, when monotonicity is established.

Example solved. For $u_n = 80(\frac{1}{2})^n$, the first ranking such as $u_n < 6$ is 4: $u_3 = 10 \geq 6$ and $u_4 = 5 < 6$. The programme begins with $n = 0, u$ = 80 and repeats division by 2 as long as $u \geq 6$.

Point of vigilance

A terminalless loop may never stop if the threshold is inaccessible; $<$ and $\leq$ are not Interchangeable.

Immediate application. For $u_n = 48(\frac{1}{2})^n$, determine the first rows where $u_n \leq 6$ and where $u_n < 6$.

Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 1.1 · Taking charge of indices

The following is considered (u_n) defined for all natural n by $u_n = 2n^2 - 3n$. The first term is therefore the level 0.

1. Calculate separately u_0, u_1 and u_4, by showing the substitution of n.

2. Express u_{n+1} according to the n, then develop and reduce this expression.

3. Calculate $u_{n+1} - u_n$. By studying his sign for the integer ranks, describe the variations in the sequence.

Exercise 1.2 · Choosing the right model

A nursery now has 100 young plants, and is considering two models to predict its future.
effective: in model A, the number increases by 15 plants per month; in model B, it increases
10 per cent per month. a_n and b_n the anticipated number of staff after n month, with $a_0 = b_0 = 100$.
Two models are studied separately.

1. **Model A.** Write the relationship between a_{n+1} and a_n. Recognize the nature of the sequence, then express a_n according to the n. Calculate a_2.

2. **Model B, after the study of geometric sequences.** Resuming the same questions for (b_n) and calculate b_2.

3. For each model, explain what is assumed to be constant.
   A very long-term forecast could become unrealistic.

Exercise 1.3 · Find a sequence

The rest (u_n) is geometric, defined from row 0. q It's strictly positive.
knows only $u_1 = 6$ and $u_3 = 54$.

1. Determine q by linking the two known terms. Present all the steps that allow
   choose its value.

2. Deduct from it $u_0$then u_5.

3. What would change the deletion of the hypothesis? $q > 0$? Give all possible reasons and
   explain whether the value u_5 would be amended.

Exercise 1.4 · Seat rows

A showroom has 12 rows. The first row has 18 seats; each row has a seat.
next has two more seats than the previous one. u_n the number of seats in the
row n, for 1 $\leq n \leq 12$.

1. Define the sequence (u_n) and calculate the number of seats in the last row.

2. Calculate the total capacity of the room. Explain the number of terms retained in the sum.

3. Calculate the average number of seats per row and use this result to control the total found.

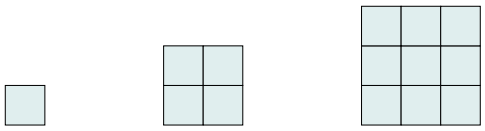
Exercise 1.5 · Accumulate squares

Patterns are built with identical square tiles. The row 1 pattern contains a tile.
To pass from the rank pattern n at the rank level n + 1, add 2 n + 1 tiles. u_n the
number of tiles in the row pattern n, for $n \geq 1$.

1. Calculate u_2 and u_3. Draw a layout of these tiles that suggests a formula for u_n.

2. Conjecture an expression of u_n according to the n.

3. Verify that this expression gives the first term and satisfies the construction rule.
   why these two audits identify the follow-up.



Rank 1 Rank 2 Rank 3

A possible provision for the first three reasons; the drawing helps to formulate a conjecture, which
We have to justify that.

Fiscal year 1.6 · Two refunds

A person must repay 600 monetary units, without interest. Two plans are envisaged.
Plan A, it pays 50 units at the end of each month until repayment.
The calculation of model B is carried out without
Rounded. a_n and b_n the remaining debts after the payment of the month n, $a_0 = b_0 = 600$.

1. Express a_n until the end of the debt. At what month is the repayment completed?

2. Express b_n. Determine the first month in which the remaining debt is strictly less than 10 units
   ; Check the previous month.

3. In Model B without rounding, does the debt become exactly zero after a finite number of
   month ? Justify and comment on the difference with a real situation.

Exercise 1.7 · Read a list

A program must retain the terms of the sequence defined by $u_0 = 2$ and $u_{n+1} = 3u_n - 1$. It
starts with the instructions `u=2` and `L=[u]`. He then executes five times, in this order, `u=3*u-1`
then `L.append(u)`.

1. Run the first two passages from the loop by hand. Then give the contents of the list.

2. How many elements contains the list at the end of the five passages? What is the rank of the last term recorded?

3. Write the complete program in Python, then check the last item value.
why the number of terms exceeds the number of passages by one unit.

Fiscal year 1.8 · Deferred review

This question is dealt with one week after the chapter, without consulting the forms.
defined for $n \geq 2$ by $u_2 = 12$ and $u_{n+1} = u_n - 3$.

1. Calculate u_7 specifying how many passages separate rows 2 and 7.

2. Calculate the sum $u_2 + u_3 + \cdots + u_7$. Explain the difference between the number of terms of this sum and the number of passages calculated on the first question.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 64, questions a and b. Calculate the first four terms.
substitutions before simplifying.

CQFD, exercise 11 p. 64, all questions. Calculate the terms successively and keep
the trace of the four stages.

MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $u_n = 3n + 2$: u_4 value
   - A. 14
   - B. 18
   - C. 12
   - D. 6
2. $u_0 = 2$, $u_{n+1} = u_n + 5$: u_3 value
   - A. 10
   - B. 12
   - C. 17
   - D. 15
3. An 8 per cent increase is the common difference / ratio
   - A. 0,08
   - B. 1,8
   - C. 0,92
   - D. 1,08
4. From u_3 to u_9 included, we count
   - A. 6
   - B. 7
   - C. 9
   - D. 12
5. $1 + 2 + \cdots +$ 20 is
   - A. 200
   - B. 420
   - C. 210
   - D. 190

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $u_n = -3 \times 2^n$ is
   - A. Strictly increasing
   - B. constant
   - C. Neither increasing nor decreasing
   - D. Strictly decreasing
2. 1 + 2 + 4 + 8 is
   - A. 15
   - B. 16
   - C. 14

D. 8

3. For $u_n = \dfrac{16}{2^n}$, first row where $u_n < 2$:

A. 2

B. 3

C. 4

D. 5

4. The future $(-1)^n$

A. tend towards 0

B. has no limit

C. is increasing

D. is constant

5. A list contains u_0 to u_{10} ; its length is

A. 10

B. 9

C. 12

D. 11

True or false?

1. Any geometric sequence of common ratio $q > 1$ is increasing.

2. If two sequences have their first three equal terms, they are equal.

3. For $q = 1$, $1 + q + \cdots + q^n = n + 1$.

Personal review

I distinguish explicit and recurring; I recognize a constant increase or rate; I count the terms; I control the previous rank of a threshold.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.

Corrected and conduct of chapter 1

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. 11: the multiplication precedes the addition. *Action:* Rewrite the steps of two calculations with parentheses.
2. $1,\ 8,\ \frac{1}{4}$. *Action:* Build the power chain by dividing by 2.
3. $80 \times 1{,}05 = 84$. *Action:* Connect 5 % to $\frac{5}{100}$ then at the coefficient of 1.05.
4. -2 and 10. *Action:* Replace x in parentheses before calculating.
5. $10 - 3 + 1 = 8$. *Action:* Write the list and find the calculation.
6. $0, 1, 2, 3$. *Action:* Draw a table with two columns, row and value; test range(1,4).

Research activity

Offer A: last 45 kg, sum $\frac{6(20+45)}{2}$ = 195 kg. B: $\frac{20(1-1{,}2^6)}{1-1{,}2} = 198{,}5984$ kg. B exceeds A in total, even if some initial terms are lower. Instrumented calculation allowed here; to argue on cumulation and not on the last term alone.

Immediate applications

1. $2, 3, 6, 11$. The formula $w_n = n^2$ +2 fits: it gives the first term and ($n+1)^2+2-(n^2+2) = 2n+1$.
2. $37 - 17 = 5r$, therefore r = 4 then $u_0 = 17 - 3 \times 4 = 5$.
3. $u_{n+1} - u_n = -4 \times 3^n < 0$: decreasing. v_n alternate 2 and -2: neither increasing nor decreasing.
4. 4 + 8 + 16 + 32 = 60, or $\frac{4(1-2^4)}{1-2} = 60$.
5. u_3 = 6: row 3 for ≤ 6, row 4 for < 6.

Exercises: approach and results

Exercise 1.1 · Take charge of the clues. $u_0 = 0, u_1 = -1, u_4$ = 20. Develop with parentheses: $u_{n+1} = 2(n+1)^2 - 3(n+1) = 2n^2 + n - 1$; difference 4 $n - 1$. The sign changes between rows 0 and 1: the sequence decreases first, then grows from 1.

Exercise 1.2 · Choose the right model. $a_n = 100 + 15n$ and $b_n = 100(1{,}1)^n$, therefore $a_2 = 130$, b_2 = 121. A assumes a constant net addition; B a constant net rate. In reality, rounding and the capacity to accommodate limit these models.

Exercise 1.3 · Find a sequence. $u_3 = q^2 u_1$ so $q^2 = 9$. Positivity requires $q = 3$, where $u_0 = 2$ and $u_5 = 486$. Without this specification, $q = -3$ would also be possible, with $u_0 = -2$ and the same $u_5 = 486$.

Exercise 1.4 · Seat rows. The last row has 18 + 11 $\times$ 2 = 40 seats. Sum $\frac{12(18+40)}{2}$ = 348. Check: average 29 seats per row, between 18 and 40.

Exercise 1.5 · Accumulate squares. We get 4 then 9 squares. A integer side square n completed by a line and column requires 2 n + 1 boxes. The formula n^2 has the right first term and $(n+1)^2 - n^2 = 2n$ + 1. The generation is the same.

Fiscal year 1.6 · Two refunds. For A: $a_n = 600 - 50n$ for 0 $\leq n \leq$ 12, then 0. For B: $b_n = \frac{600}{2^n}$. $b_5 = 18{,}75$ and $b_6 = 9{,}375$; strict threshold at row 6. B never gives exactly 0 in this ideal model: do not announce a finished refund.

Exercise 1.7 · Read a list. The first three elements are 2, 5, 14. An initial value and five additions give six elements; indices 0 to 5. $u_5 = 365$.

Exercise 1.8 · Deferred review. $u_7 = 12 + 5(-3) = -3$. Six terms: $\frac{6(12-3)}{2}$ = 27. count the passages, the 6 terms.

CQFD, exercise 9 p. 64, questions a and b. a : $-1, 0, 3, 8$. b : $2, \frac{7}{2}, 4, \frac{17}{4}$. The first row is 0 ; the denominator of b is n + 1 and does not cancel on $\mathbb{N}$.

CQFD, exercise 11 p. 64, all questions. From u_0 = 2, the rule $u_{n+1} = 2u_n - 1$ gives $u_1 = 3$, $u_2 = 5$, $u_3 = 9$, u_4 = 17. The rule uses the previous term, not its rank.

MCQ: answers and reasons

MCQ 1.

1. A. $3 \times 4 + 2 = 14$.

2. C. Three additions: 2 + 3 $\times$ 5 $= 17$.

3. D. The coefficient is 1 + $\frac{8}{100}$.

4. B. $9 - 3 + 1 = 7$.

5. C. $\frac{20 \times 21}{2} = 210$.

MCQ 2.

1. D. The difference is $-3 \times 2^n < 0$.

2. A. Four terms; $\frac{1 - 2^4}{1 - 2} = 15$.

3. C. $u_3 = 2$, $u_4 = 1$.

4. B. The terms alternate between two distinct values.

5. D. Eleven rows from 0 to 10.

True or false

1. False: $u_n = -2^n$ is strictly decreasing.

2. False: $u_n = 0$ and $v_n = n(n-1)(n-2)$ coincide with ranks 0.12, but not with rank 3.

3. True: each term is 1 and there are n+1 terms.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.